Category Theory for Mathematical Physics — Exercise Sheet 6

F-symbols and the pentagon

Variant: Hints and answers

Conventions as in Lecture 6: the categories are multiplicity-free, with vertices ψ_c^{ab}; the F-matrix F_d^{abc} has rows e and columns f in the order of the list of simple objects (Definition 6.1); an F-symbol with a unit leg is 1 when its trees exist, and every symbol whose trees do not exist is 0. Fibonacci has $1, \tau$ and Ising $1, \sigma, \psi$ (the bare ψ is Ising's fermion; a vertex always carries its labels). $\mathbb{Z}_2 = \{1, s\}$ is written multiplicatively, as in Lecture 3. The gauge numbers are u_c^{ab} (Theorem 6.4); $\varphi = (1+\sqrt{5})/2$; H is the Hadamard matrix; x, A, B, C, D are the Fibonacci unknowns of Theorem 6.6. In Exercise 5, $\uparrow, \downarrow$ are the spin-$\frac{1}{2}$ states $m = \pm\frac{1}{2}$. In Exercise 6, $\gamma_1, \gamma_2, \gamma_3$ name three F-symbols (the labels g of the pentagon and the group elements g of Lecture 3 are other letters).

Exercise 1 (Boson and semion). Take the fusion ring of $\mathbb{Z}_2$: $s \otimes s \cong 1$.

(i) Show that the only F-symbol without a unit leg is $x = [F_s^{sss}]_{11}$, and that the pentagon equation of Theorem 6.3 reduces to $x^2 = 1$.

(ii) Show that x is gauge invariant, and conclude that the fusion ring has exactly two solutions up to gauge. Which categories of Lecture 3 are they, and what is κ_s for each?

(iii) Draw the F-move Equation (6.2) for $g = h = k = s$.

Hint. (i) In $[F_d^{abc}]_{ef}$ the labels e, f, d are forced by a, b, c. (ii) (6.4), with $d_s = 1$.

Answer. (i) The instance $a = b = c = d = s$ reads $1 = x \cdot 1 \cdot x$; every other instance reads $1 = 1$ or $x = x$. (ii) Its factor is $u_s^{s1}\, u_1^{ss} / (u_1^{ss}\, u_s^{1s}) = 1$; $x = 1$ is $\mathrm{Vec}_{\mathbb{Z}_2}$ (the boson, $\kappa_s = 1$), $x = -1$ the semion ($\kappa_s = -1$). (iii) The left tree with inner label 1 equals $\omega(s,s,s) = x$ times the right tree with inner label 1.

Exercise 2 (Fibonacci in terms of d_τ, and Yang–Lee). Let d_τ be a root of $d^2 = 1 + d$, and let $d_\tau^{-1/2}$ be a square root of d_τ^{-1}.

(i) Show that $x = 1$ and

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} d_\tau^{-1} & d_\tau^{-1/2} \\ d_\tau^{-1/2} & -d_\tau^{-1} \end{pmatrix}$$

satisfy the six equations listed before Theorem 6.6, and that $d_\tau = \varphi$ gives the unitary solution there.

(ii) For $d_\tau = \varphi$, write the inverse F-move Equation (6.3) for three τ's with total charge τ whose last two fuse to 1.

(iii) For $d_\tau = -1/\varphi$ (Yang–Lee) write the matrix, and show that it squares to the identity but is not unitary. Compute $\kappa_\tau = d_\tau A$ and $1 + d_\tau^2$ for both roots, and compare the second with $\dim \mathrm{Fib} = 2 + \varphi$ under $\sqrt{5} \mapsto -\sqrt{5}$. Why is Yang–Lee no anyon model?

Hint. (i) Write d for d_τ and use $d^2 = 1 + d$ to clear d^{-2}. (iii) The other root of $d^2 = 1 + d$ is $1 - \varphi = -1/\varphi$.

Answer. (i) $A = d^{-1}$, $BC = d^{-1}$, $D = -d^{-1}$, $x = 1$: the equations become $d^{-1} + d^{-2} = 1$ and identities. (ii) $R_1 = \varphi^{-1} L_1 + \varphi^{-1/2} L_\tau$. (iii) $\begin{pmatrix} -\varphi & i\sqrt{\varphi} \\ i\sqrt{\varphi} & \varphi \end{pmatrix}$, entries of modulus > 1; $\kappa_\tau = 1$ for

both; $1+d^2 = 2+\varphi$ and $3-\varphi$, Galois conjugates; not unitary in any gauge, since $|A| = \varphi > 1$ is gauge invariant.

Exercise 3 (What a gauge can change). (i) For Fibonacci, with $u_1^{\tau\tau} = p$ and $u_\tau^{\tau\tau} = q$, compute from Theorem 6.4 the factors on x, A, B, C and D. Show that BC is invariant and B/C is multiplied by q^4/p^2. The numbers $x = 1$, $A = \varphi^{-1}$, $B = 1$, $C = \varphi^{-1}$, $D = -\varphi^{-1}$ solve the pentagon; can a gauge with $|u| = 1$ make them unitary?

(ii) For Ising, show that $[F_\sigma^{\sigma\sigma\sigma}]_{11}$ is invariant, and compute $F_\sigma^{\sigma\sigma\sigma}$ and $[F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma}$ after the gauge $u_\psi^{\sigma\sigma} = i$, all other u equal to 1. Is the new matrix unitary?

Hint. (i) $[F_\tau^{\tau\tau\tau}]_{ef}$ has the factor $u_\tau^{\tau f}u_f^{\tau\tau}/(u_e^{\tau\tau}u_\tau^{e\tau})$, with $u_\tau^{1\tau} = u_\tau^{\tau 1} = 1$. (ii) The same formula with σ in place of τ.

Answer. (i) x, A, D: factor 1; B: q^2/p; C: p/q^2. No: $|B/C| = \varphi$ cannot change when $|u| = 1$, and a unitary solution has $|B| = |C|$. (ii) $\frac{\kappa}{\sqrt{2}}\begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix}$, unitary; $[F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma} = -1$ unchanged.

Exercise 4 (Pentagon instances by hand). (i) Write out the pentagon equation of Theorem 6.3 for Fibonacci with $a = b = c = d = e = g = k = \tau$ and $f = l = 1$, and show that it is $A^2 + BCD = 0$. Draw the five trees of the pentagon for these labels.

(ii) Find labels whose instance is $BD = xBD$.

(iii) For Ising, with $a = b = c = d = g = k = \sigma$ and $e = f = l = 1$, show that the instance reads

$$1 = \tfrac{\kappa^2}{2}\left(1 + [F_1^{\sigma\psi\sigma}]_{\sigma\sigma}\right)$$

when $F_\sigma^{\sigma\sigma\sigma} = \kappa H$, and check it for the data of Theorem 6.7.

Hint. (i) A symbol whose trees do not exist is 0. (ii) Put $g = 1$.

Answer. (i) Left side 0 (no right tree for $F_\tau^{1\tau\tau}$ with channel 1); right side $A\cdot 1\cdot A + B\cdot D\cdot C$. (ii) $a = b = c = d = e = f = k = l = \tau$, $g = 1$. (iii) $h = 1$ gives $\kappa^2/2$ and $h = \psi$ gives $(\kappa^2/2)[F_1^{\sigma\psi\sigma}]_{\sigma\sigma}$; with $[F_1^{\sigma\psi\sigma}]_{\sigma\sigma} = 1$ the instance is $1 = \kappa^2$.

Exercise 5 (SU(2) recoupling is a $6j$ symbol). In $\mathrm{Rep}(\mathrm{SU}(2))$ (semisimple, rigid and multiplicity-free, though not finite) the F-symbols are defined as in Definition 6.1. Take isometric vertices given by the Clebsch–Gordan coefficients:

$$\psi_0^{\frac12\frac12}(1) = \tfrac{1}{\sqrt2}(\uparrow\downarrow - \downarrow\uparrow),$$

$$\psi_1^{\frac12\frac12}\colon\ |1,1\rangle \mapsto \uparrow\uparrow,\ |1,0\rangle \mapsto \tfrac{1}{\sqrt2}(\uparrow\downarrow + \downarrow\uparrow),$$

$$\psi_{\frac12}^{1\frac12}\colon\ \uparrow \mapsto \sqrt{\tfrac23}\,|1,1\rangle\downarrow - \sqrt{\tfrac13}\,|1,0\rangle\uparrow,$$

$$\psi_{\frac12}^{\frac121}\colon\ \uparrow \mapsto \sqrt{\tfrac13}\uparrow|1,0\rangle - \sqrt{\tfrac23}\downarrow|1,1\rangle,$$

and $\psi_{\frac12}^{0\frac12} = \psi_{\frac12}^{\frac120} = \mathrm{id}$.

(i) Explain why rescaling the vertices to the normalisation Equation (5.1) does not change the F-symbols. Compute the trees L_0, L_1, R_0, R_1 for three spin-$\frac12$ with total spin $\frac12$ on the state $\uparrow$, and the matrix $F_{\frac12}^{\frac12\frac12\frac12}$.

(ii) Compare with the Racah–Wigner formula

$$[F_d^{abc}]_{ef} = (-1)^{a+b+c+d}\sqrt{(2e+1)(2f+1)}\begin{Bmatrix} a & b & e \\ c & d & f \end{Bmatrix},$$

given that the $6j$ symbols $\left\{\begin{smallmatrix} 1/2 & 1/2 & e \\ 1/2 & 1/2 & f \end{smallmatrix}\right\}$ are $-\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{6}$ for $(e,f) = (0,0), (0,1), (1,0), (1,1)$. Compute $d_{1/2}[F^{\frac{1}{2}\frac{1}{2}\frac{1}{2}}_{\frac{1}{2}}]_{00}$ with $d_{1/2} = 2$.

(iii) Write the pentagon instances with $a = b = c = d = \frac{1}{2}$, $e = 0$, $g = k = \frac{1}{2}$ (one for each f, l). Given that the 1×1 symbols in them are 1, show that together they say $(F^{\frac{1}{2}\frac{1}{2}\frac{1}{2}}_{\frac{1}{2}})^2 = 1$. (This instance is degenerate: (i) already implies it. The general pentagon for these F-symbols is the Biedenharn–Elliott identity, which the check script verifies on representations built with rep-engine.)

Hint. (i) Look at how a rescaling of the vertices changes L_e and R_f, with the bubble. (iii) Which channels f, l, h are allowed with $e = 0$?

Answer. (i) Both families scale by the common factor $(d_a d_b d_c/d_d)^{1/4}$; $F^{\frac{1}{2}\frac{1}{2}\frac{1}{2}}_{\frac{1}{2}} = \begin{pmatrix} -1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix}$. (ii) The formula gives the same matrix; $d_{1/2}F_{00} = -1 = \kappa_{1/2}$, as in Lecture 4. (iii) $\delta_{fl} = \sum_h F_{fh}F_{hl}$, true as F is real, symmetric and orthogonal.

Exercise 6 ⋆. *The Ising F-symbols, and why there are no others.* Name the Ising F-symbols without a unit leg as follows:

$$
\begin{gathered}
n_1 = [F^{\psi\psi\psi}_{\psi}]_{11},\ n_2 = [F^{\psi\psi\sigma}_{\sigma}]_{1\sigma}, \\
n_3 = [F^{\psi\sigma\sigma}_{1}]_{\sigma\psi},\ n_4 = [F^{\psi\sigma\sigma}_{\psi}]_{\sigma 1}, \\
n_5 = [F^{\psi\sigma\psi}_{\sigma}]_{\sigma\sigma},\ n_6 = [F^{\sigma\psi\sigma}_{\psi}]_{\sigma\sigma},\ n_7 = [F^{\sigma\sigma\psi}_{1}]_{\psi\sigma}, \\
\gamma_1 = [F^{\sigma\psi\sigma}_{1}]_{\sigma\sigma},\ \gamma_2 = [F^{\sigma\psi\psi}_{\sigma}]_{\sigma 1}, \\
\gamma_3 = [F^{\sigma\sigma\psi}_{\psi}]_{1\sigma},
\end{gathered}
$$

and $F^{\sigma\sigma\sigma}_{\sigma} = (m_{ef})$ with $e, f \in \{1, \psi\}$.

(i) Compute the gauge factors of $\gamma_1, \gamma_2, \gamma_3$, and show that every solution of the pentagon with invertible F-matrices (as for any category) is gauge equivalent to one with $\gamma_1 = \gamma_2 = \gamma_3 = 1$.

(ii) With $\gamma_1 = \gamma_2 = \gamma_3 = 1$ the following are pentagon instances:

$$
\begin{gathered}
n_1 = n_7, \quad n_1 n_2 = n_2, \quad n_2 = 1, \\
n_6^2 = n_2, \quad n_2 n_5^2 = n_2, \quad n_1 = n_2 n_3 n_4, \quad n_3 n_7 = n_4, \\
m_{1\psi} = m_{11}, \quad m_{\psi 1} = n_4 m_{11}, \\
m_{1\psi} m_{\psi 1} + m_{11}^2 = 1, \quad n_6 m_{1\psi} m_{\psi 1} + m_{11}^2 = 0, \\
n_5 m_{\psi\psi} = n_4 m_{1\psi}, \quad n_5 m_{\psi 1} = n_7 m_{\psi\psi}, \\
m_{1\psi} m_{\psi\psi} + m_{11} m_{1\psi} = 0.
\end{gathered}
$$

Derive the last one from the labels $a = b = c = d = g = k = \sigma$, $e = f = 1$, $l = \psi$. Then show that these fourteen equations have exactly two solutions: $F^{\sigma\sigma\sigma}_{\sigma} = \kappa H$ with $\kappa = \pm 1$, $n_5 = n_6 = -1$, and every other symbol 1.

(iii) Conclude that the Ising fusion ring carries exactly two fusion categories up to gauge, and that they are told apart by κ.

Hint. (i) The three factors involve $u^{\psi\sigma}_{\sigma}$, $u^{\psi\psi}_{1}$ and $u^{\sigma\sigma}_{\psi}$ respectively. (ii) Find $m_{11} \neq 0$ first; the step that needs care is the sign of n_4.

Answer. (i) $u_\sigma^{\psi\sigma}/u_\sigma^{\sigma\psi}$, $u_1^{\psi\psi}/(u_\sigma^{\sigma\psi})^2$, $u_\sigma^{\sigma\psi}u_\psi^{\sigma\sigma}/u_1^{\sigma\sigma}$: set $u_\sigma^{\sigma\psi} = u_1^{\sigma\sigma} = 1$ and solve for the other three. (ii) $n_1 = n_2 = n_3 = n_4 = n_7 = 1$, $n_5 = n_6 = -1$, $m_{11} = m_{1\psi} = m_{\psi 1} = -m_{\psi\psi} = \kappa/\sqrt{2}$. (iii) Every solution is gauge equivalent to one of these two, and $\kappa = d_\sigma m_{11}$ is invariant.